Bipolaris-RNAi Designer (BRD) v2.0.0 — Audited & Corrected Research Report

Generated 2026-11-11 04:45 UTC | Audit directory: C:\Users\ASUS BANGLADESH\Desktop\New folder (2)\Output\11_post_execution_audit

1. Audit Scope & Version Governance

Bipolaris-RNAi Designer (BRD) v2.0.0 Post-Execution Audit (2026-11-11 04:45 UTC). Original v1 outputs in `C:\Users\ASUS BANGLADESH\Desktop\New folder (2)\Output\10_final_reports` remain strictly preserved. All corrected deliverables reside in `C:\Users\ASUS BANGLADESH\Desktop\New folder (2)\Output\11_post_execution_audit`.

2. What Was Executed, Substituted & Incomplete

Software & Component Status:

3. Target Gene Verification & Evidence Audit

- **CsPKS1 GenBank Audit (`BRD_CsPKS1_Record_Audit_v2.tsv`):** `HQ830033.1` (`6,862 nt` genomic clone containing `6,462 nt` complete CDS, `100.0%` match to `COCSADRAFT_157080`) plus 8 partial CDS isolate amplicon records `KF769134.1`–`KF769141.1` (`441–716 nt`, `100.0%` identity over covered span). - **Published Primer Verification (`BRD_Target_Primer_Alignments_v2.tsv`):** `CsFUS3` 3/3 exact in `ND90Pr`; `CsSTE11` 3/4 exact in `ND90Pr` + internal `CsSTE11-F6` 19/20 (`1` synonymous SNP) in `ND90Pr` & 20/20 (`100%`) exact in `ND93-1`; `CsSp1` 5/5 `100%` exact Zhang et al. (2022) primers in both `ND90Pr` and `LK93` (including exon 1–2 splice-junction spanning `CsSP1-QRT2-R`). Complete details and primer-vs-standalone-accession distinction are in `BRD_Target_Gene_Identity_Evidence_v2.tsv`.

4. Corrected Isolate Conservation & Denominators

All **19/19** biological isolates contain intact homolog loci (`0` gene absences or unresolved sequences). The 2/19 partial/frameshifted CDS records for `CsPKS1`, `CsFUS3`, and `CsSTE11` occur solely in unpolished long-read draft assemblies (`WAI2406` and `WAI2411`) with internal homopolymer indels.

5. Corrected Off-Target Classification & Filtering Funnel (Axes A, B & C)

Full k-mer screening across 7 non-target RefSeq transcriptomes (221,398 wheat transcripts, 63,294 barley transcripts, 5 beneficial fungi) and `ND90Pr` non-self paralogs identified **16 candidates** with exact 21-mer beneficial-fungi matches (`12/13` `CsFUS3` including v1 Pareto lead `BRD_CsFUS3_CAND013`, plus `4/14` `CsSTE11`: `CAND001, CAND002, CAND003, CAND009`), compared with **8** caught in v1. Separating random 19-nt 1-mismatch noise (`E=1.8–7.9`) from true `20/21` near-guides (`452` BLAST rows) and `>= 30 nt` alignments (`286` rows across 16 candidates) eliminated the v1 collapse where all 50 candidates appeared either 'High-similarity' or 'Exact'.

v2 Off-Target Tier Counts:

Sequential Candidate Filtering Funnel (`off_target_filtering_funnel_v2.tsv`):

6. Synchronised Candidate Tables & Recalculated Rankings (v2)

All 50 rows in `BRD_dsRNA_Candidates_v2.tsv` were synchronised (zero stale `PENDING_M08` fields; 100% sequence/FASTA/GC/length/conservation agreement). Under v2 filters (0 exact 21-mers in wheat, barley, beneficial fungi, or fungal paralogs; GC 30–65%; max homopolymer <=5; conservation >=85%), **33/50 candidates passed** (**2 High**, **31 Moderate**, **0 Low priority**, and **17 Excluded**).

Why Final Rankings Changed: 1. `BRD_CsFUS3_CAND013` (v1 `#13` / Pareto Rank 1) and `CsFUS3` candidates `CAND005–009` were **excluded** after exposing 3–5 exact 21-mer matches to beneficial fungal MAPKs (`BRD_CsFUS3_CAND013` dropped to `#40`, `safety_v2=0.02`). 2. `BRD_CsSTE11_CAND012` strengthened at **#1** (score

`0.7832 -> 0.917`, `safety_v2 = 0.915`, `100.0%` conservation, `0` wheat/barley `20/21`, `0` >=30 nt alignments, **Rank #1 across 1,000-bootstrap and all 5 sensitivity scenarios**), joined by `BRD_CsSTE11_CAND013` (**#2**, `safety_v2 = 0.875`, score `0.9059`, High priority) and `BRD_CsSTE11_CAND005` (**#3**, score `0.8902`, Moderate priority). 3. `BRD_CsPKS1_CAND013` rose from v1 rank **#26** to v2 **Pareto Rank 1 (#9 overall, safety_v2 = 0.965 — cleanest sequence specificity in the entire panel, score = 0.7957)** once spurious 19nt-1mm penalties were removed.

Top 10 Computationally Passing Candidates (v2):

7. Evidence-Based Biological Target Scoring & Expression Audit

v1 hardcoded `bio_eval_meta` scores (`0.98/0.96/0.86/0.62`) and arbitrary `min(1.0, RPKM/150.0)` were replaced with an auditable 4-criterion biological rubric (`biological_target_scoring_rubric_v2.json`; weights `0.35 ExpPatho + 0.20 CompDB_IPR + 0.25 IsoCons + 0.20 VerifConf`) and an explicit distribution audit of `GSE130517` (`12,034` genes x `139` columns).

Programmatic Biological Target Prioritisation (`BRD_Target_Biological_Prioritisation_v2.tsv`):

Per-Criterion Evidence Records (`BRD_Biological_Scoring_Evidence_Matrix_v2.tsv`):

`GSE130517` Expression Distribution Audit & Exclusion from Composite Score (`GSE130517_expression_distribution_audit_v2.tsv`):

- **Why `min(1.0, RPKM/150.0)` was excluded:** Early/mock samples (`1A`, `2A`, `2B`, `3A`, `4A`) contain only `1,921–3,808` total mapped fungal reads across `148–213` genes, creating extreme 1-read RPKM artifacts (v1 `CsSp1` peak `1438.44` came from **1 single raw read** in `3A_I (raw_reads=1)`; v1 `CsSTE11` peak `87.63` came from **1 single raw read** in `2B_III (raw_reads=1)`). In valid colonized root samples (`3B/4B/5B`, `107,929–578,376` fungal reads), mean RPKMs are `79.3` (`CsSp1`), `62.53` (`CsFUS3`), `22.12` (`CsSTE11`), and `13.19` (`CsPKS1`). Expression is therefore retained strictly as separate qualitative/quantitative context (`expression_excluded_from_composite = True`).

8. Separate Biological Target vs dsRNA Candidate Rankings

Biological importance as a pathogenicity gene and suitability as an exogenous dsRNA target diverge cleanly across the four loci:

- **Biological Target Ranking:** #1 CsFUS3 (0.9915), #2 CsSTE11 (0.9866), #3 CsSp1 (0.6993), #4 CsPKS1 (0.6913). `CsFUS3` and `CsSTE11` lead on curated PHI-base/DEG essentiality, InterPro domain depth, RefSeq/UniProt traceability, and published pathogenicity knockouts, whereas `CsSp1` lacks PHI-base/DEG/InterPro depth and `CsPKS1` (`PHI:2260`) retains unaffected barley virulence on knockout. - **dsRNA Candidate Targetability Ranking:** #1 CsSTE11 (0.917), #2 CsPKS1 (0.7988), #3 CsSp1 (0.7911), #4 CsFUS3 (0.7259). `CsSTE11` (`2,880 nt`) ranks #1 on targetability (non-conserved region candidates `CAND012`/`CAND013`); `CsPKS1` ranks #2 overall (`CAND010`, `0.7988`, with `CsPKS1_CAND013` achieving the single cleanest off-target score `safety_v2=0.965`); `CsSp1` ranks #3 (`0.7911`, `0` exact 21-mers/long hits, moderate Wheat `20/21` caution); `CsFUS3` ranks #4 for exogenous dsRNA despite ranking #1 biologically (`12/13` exact beneficial-fungi 21-mers, remaining `CAND003` shares a `126 nt` 85% fungal MAPK block at `safety_v2=0.445`).

9. Scientific Boundaries & Validation Prerequisites

1. Passing sequence off-target screening (`no match detected` or `minimal 19/21 signal`) does **not** prove biological safety. 2. Endogenous hairpin-RNAi knockdown of `CsPKS1` (Leng et al., 2011) does **not** demonstrate exogenous dsRNA uptake or SIGS efficacy. 3. Independent test execution logs (`pytest tests/test_audit_v2.py -v`) are recorded in `BRD_Test_Results_v2.txt`. 4. Priority wet-lab progression: verify exogenous Cy3/FAM-dsRNA uptake in *B. sorokiniana* conidia/hyphae -> qRT-PCR knockdown of `CsSTE11` (`BRD_CsSTE11_CAND012`) & `CsSp1` (`BRD_CsSp1_CAND008`) -> detached leaf infection assays with non-targeting/scrambled dsRNA controls and wheat/barley off-target RT-qPCR verification (checking predicted `20/21` sites for `CsSp1`).

Evidence-Based Biological Target Prioritisation (v2)

t_rank	target	score_pathogenicity_experimental	score_comparative_pathogenicity_essentiality	score_isolate_conservation	score_verification_confidence	biological_target_score	evidence_confi
	CsFUS3	1.0	1.0	0.966	1.0	0.9915	High (multi-sou
	CsSTE11	1.0	1.0	0.9462	1.0	0.9866	High (multi-sou
	CsSp1	1.0	0.0	0.9973	0.5	0.6993	Moderate (prim
	CsPKS1	0.3333	0.6667	0.9653	1.0	0.6913	High (multi-sou

Dual Target Rankings: Biological vs dsRNA Candidate Priority (v2)

target	biological_target_rank	biological_target_score	dsrna_targetability_rank	best_passing_candidate_id	best_passing_candidate_score_v2	candidates_passing_v2_filters
CsFUS3	1	0.9915	4	BRD_CsFUS3_CAND003	0.7259	1
CsSTE11	2	0.9866	1	BRD_CsSTE11_CAND012	0.917	9
CsSp1	3	0.6993	3	BRD_CsSp1_CAND008	0.7911	10
CsPKS1	4	0.6913	2	BRD_CsPKS1_CAND010	0.7988	13

GSE130517 Expression Distribution Audit (Excluded from Composite)

target	v1_reported_inplanta_max_rpkm	v1_peak_source_sample	valid_3B_4B_5B_raw_reads_sum	valid_3B_4B_5B_inplanta_mean_rpkm	axenic_Bs_mean_rpkm
CsPKS1	27.17	3B_II (raw_reads=50)	223	13.19	20.55
CsFUS3	90.21	3B_I (raw_reads=66)	216	62.53	210.59
CsSTE11	87.63	2B_III (raw_reads=1)	215	22.12	35.78
CsSp1	1438.44	3A_I (raw_reads=1)	75	79.3	7.85

Sequential Off-Target Filtering Funnel (v2)

filter_step	candidates_passing_individually	candidates_passing_cumulatively
0_Total_Unique_Designed_Cand	50	50
1_Sequence_QC_Length_GC_Homo	50	50
2_Zero_Fungal_Paralog_Exact_	48	48
3_Zero_Wheat_Barley_Exact_21	50	48
4_Zero_Beneficial_Fungi_Exac	34	34
5_Conservation_ge85pct_Hard_	48	33
6_Caution_No_Long_ge30nt_Hom	34	32
7_Caution_Low_Host_20of21_Ne	23	13

Corrected Isolate Conservation Denominators (v2)

s_searched	isolates_with_detected_homolog	isolates_with_complete_cds	mean_pairwise_nt_id_vs_ref_all18_pct	mean_pairwise_nt_id_vs_ref_complete_only_pct	mean_21mer_isolate_c
	19	17	99.958	99.968	98.973
	19	17	99.984	99.994	99.205

s_searched	isolates_with_detected_homolog	isolates_with_complete_cds	mean_pairwise_nt_id_vs_ref_all18_pct	mean_pairwise_nt_id_vs_ref_complete_only_pct	mean_21mer_isolate_c
	19	17	99.657	99.664	92.602
	19	19	99.959	99.959	99.116

Top 15 Ranked dsRNA Candidates (v2)

final_rank_v2	v1_final_rank	candidate_id	target	off_target_safety_score_v2	composite_score_v2	computational_priority_category_v2
1	1	BRD_CsSTE11_CAND012	CsSTE11	0.915	0.917	High computational priority
2	12	BRD_CsSTE11_CAND013	CsSTE11	0.875	0.9059	High computational priority
3	14	BRD_CsSTE11_CAND005	CsSTE11	0.845	0.8902	Moderate computational prior
4	15	BRD_CsSTE11_CAND006	CsSTE11	0.845	0.8858	Moderate computational prior
5	16	BRD_CsSTE11_CAND007	CsSTE11	0.85	0.884	Moderate computational prior
6	17	BRD_CsSTE11_CAND008	CsSTE11	0.85	0.8793	Moderate computational prior
7	28	BRD_CsPKS1_CAND010	CsPKS1	0.81	0.7988	Moderate computational prior
8	23	BRD_CsSTE11_CAND014	CsSTE11	0.745	0.7973	Moderate computational prior
9	26	BRD_CsPKS1_CAND013	CsPKS1	0.965	0.7957	Moderate computational prior
10	31	BRD_CsPKS1_CAND001	CsPKS1	0.8	0.7946	Moderate computational prior
11	33	BRD_CsPKS1_CAND011	CsPKS1	0.8	0.7928	Moderate computational prior
12	29	BRD_CsPKS1_CAND002	CsPKS1	0.79	0.7928	Moderate computational prior
13	2	BRD_CsSp1_CAND008	CsSp1	0.695	0.7911	Moderate computational prior
14	37	BRD_CsPKS1_CAND008	CsPKS1	0.78	0.7839	Moderate computational prior
15	34	BRD_CsPKS1_CAND007	CsPKS1	0.775	0.7835	Moderate computational prior

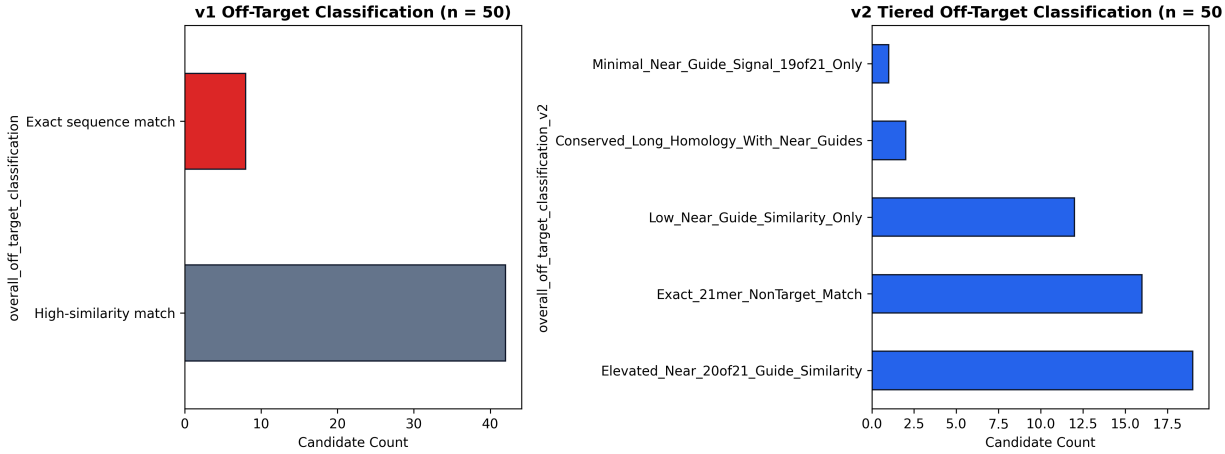


Figure 1. v1 vs v2 Off-Target Classification.

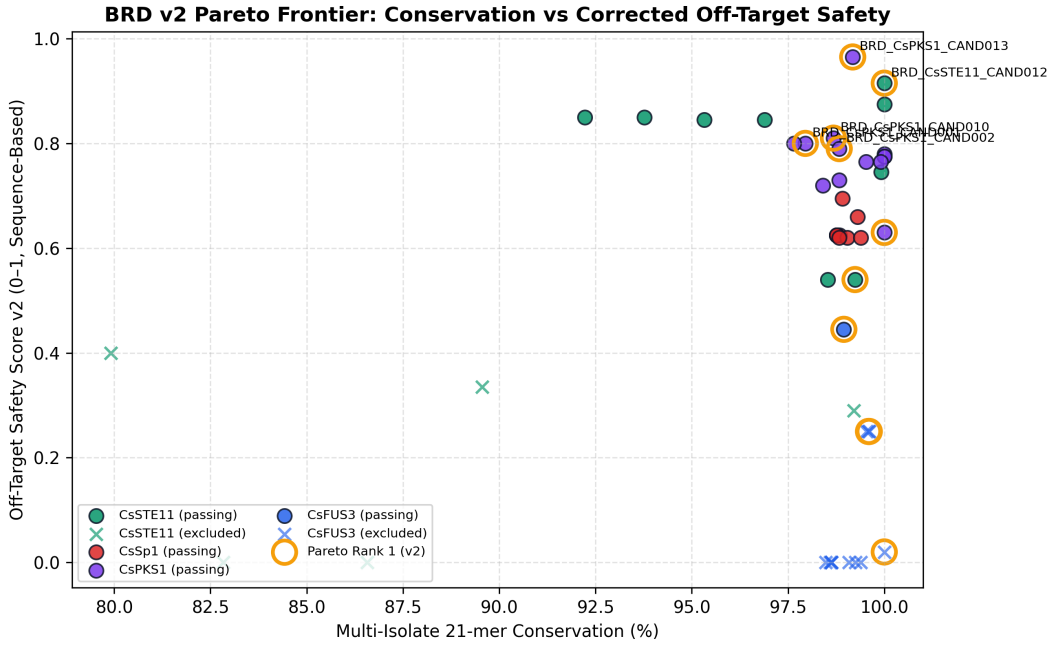


Figure 2. v2 Pareto Frontier.

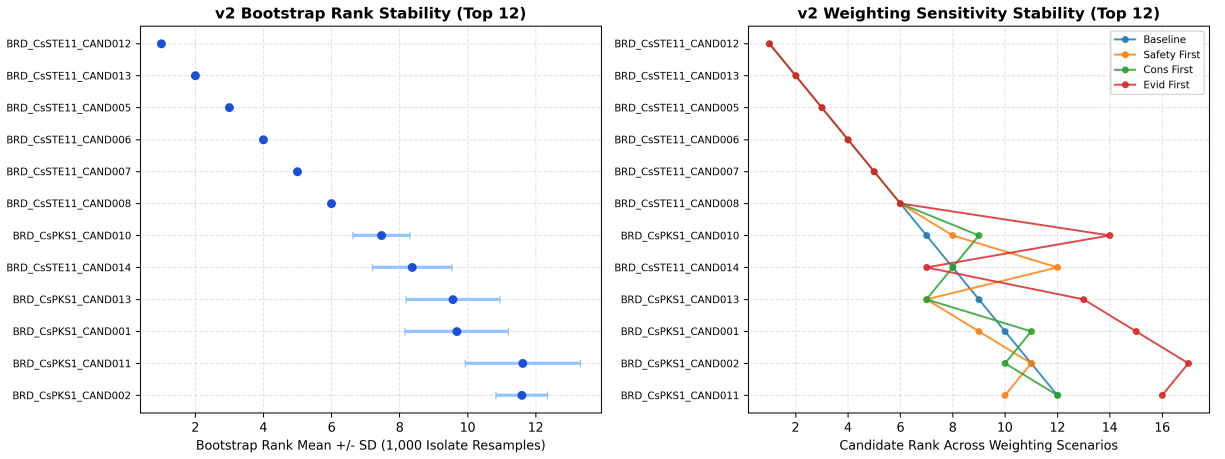


Figure 3. v2 Bootstrap & Sensitivity Analysis.

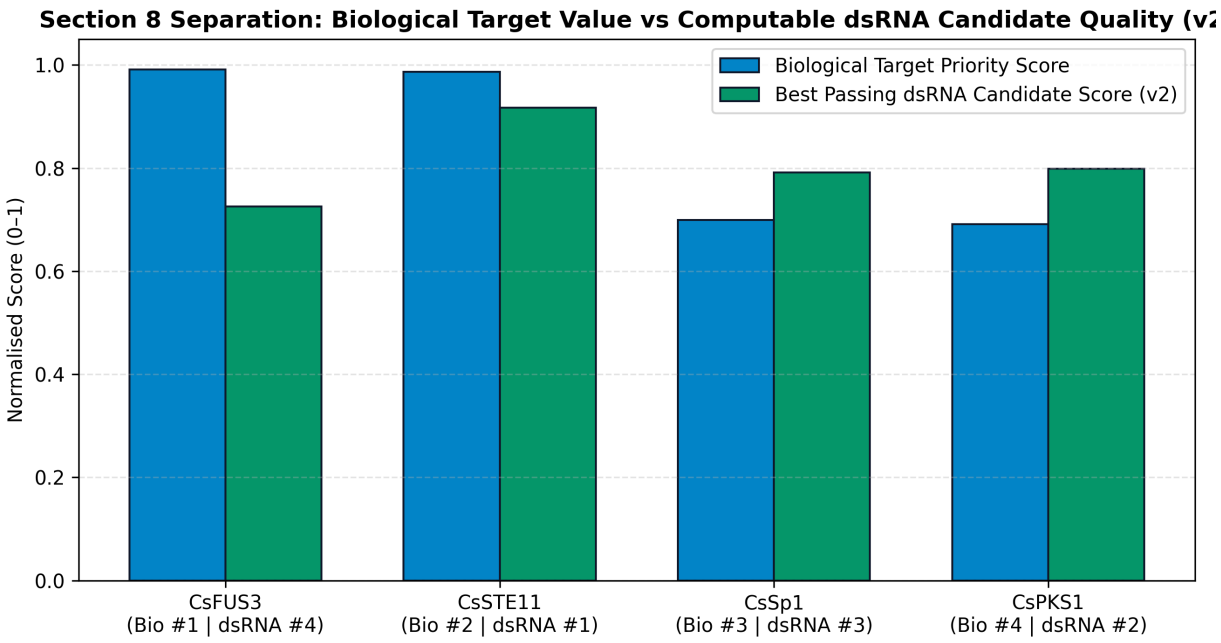


Figure 4. Biological vs dsRNA Candidate Target Rankings.